

ANALYSTLAB AFRICA CONSULTING

BUSINESS ANALYTICS REPORT

**ABC Communications Ltd
Customer Churn & Retention Analysis**

Prepared by: Nontethelelo Ivy Macebo

Role: Junior Data Analyst

August 2026

Executive Summary

ABC Communications Ltd is experiencing a significant customer-retention challenge, with 26.54% of customers having churned. The analysis identified several customer characteristics associated with increased churn, particularly contract type, tenure, internet service and payment method.

Month-to-month customers represent 55.02% of the customer base and have the highest contract-level churn rate at 43%, compared with 11% for one-year contracts and 3% for two-year contracts. Early-tenure customers are also particularly vulnerable, with customers in their first 12 months recording a 48.28% churn rate compared with only 6.68% among customers with 60-72 months of tenure. Fiber optic customers and electronic-check users also demonstrate elevated churn rates of 42% and 45%, respectively.

The analysis further found that churned customers have higher average monthly charges (R76.28) than customers who remained (R60.81). However, the correlation between monthly charges and churn is weak (0.1934), indicating that higher charges alone are unlikely to explain customer churn.

Based on these findings, ABC Communications should prioritise early-tenure retention, targeted conversion of suitable month-to-month customers to longer contracts, investigation of fiber-related customer experience, and intervention among electronic-check users. A data-driven churn-risk dashboard should also be implemented to monitor high-risk customer segments and measure the effectiveness of retention initiatives.

1. Business Understanding

1.1 Business Problem

ABC Communications Ltd commissioned the analysis to understand customer churn and identify customer characteristics and segments associated with higher levels of customer loss. The objective is to translate customer-level data into practical retention actions.

1.2 Business Questions

What does the customer base look like?

Which customer segments have the highest churn?

Does contract type influence retention?

Does tenure affect customer loyalty?

Which internet service characteristics are associated with churn?

Which payment methods have higher observed churn?

What practical actions should management take to improve customer retention?

1.3 Business Importance

Customer churn can reduce recurring revenue and increase the operational effort required to replace lost customers. A segment-based churn analysis allows management to prioritise retention resources toward customer groups showing materially higher observed churn, rather than applying the same intervention to the entire customer base.

2. Dataset Inspection

2.1 Dataset Overview

The analysis used the Telco Customer Churn dataset supplied for the case study. The working dataset contains customer identifiers, demographic attributes, tenure, service information, contract and payment details, monthly charges, total charges and churn status. The analytical work used 7,043 customer records.

2.2 Data Quality Checks

Check	Finding / Treatment
Rows / observations	7,043 customer records used in the analysis.
Columns / variables	Customer, demographic, service, contract, payment, charge and churn fields were inspected.
Missing / blank values	Blank-looking values were investigated, particularly in TotalCharges.
TotalCharges data type	Values were not consistently recognised by Excel as numeric; a cleaned numeric analytical field was used.
MonthlyCharges data type	Values were initially stored as text; MonthlyCharges_Clean was created for analysis.
Duplicate customers	Customer IDs were checked as part of data inspection.
Churn	Yes/No was retained; Churn_Numeric was created as 1 = churned and 0 = stayed for correlation.
Text quality	Extra spaces and inconsistent text values were inspected during cleaning.

2.3 Data Preparation

The raw data were preserved while cleaned analytical fields were created. MonthlyCharges_Clean and TotalCharges_Clean were used where numeric analysis required consistent numerical formats. Churn_Numeric converted the categorical churn outcome into a binary numeric variable for correlation analysis.

3. Business Data Analysis

The Excel workbook contains the required visualisations. The following results and interpretations summarise the evidence generated from the PivotTables, histograms, box plot and correlation heatmap.

3.1 Customer Base by Contract Type

Pie chart: Month-to-month customers represent 55.02% of the customer base, followed by two-year customers at 24.07% and one-year customers at 20.91%.

3.2 Overall Customer Churn

Pie chart: 73.46% of customers remained with ABC Communications while 26.54% churned.

3.3 Churn by Contract Type

Bar chart: Month-to-month customers have a 43% churn rate, compared with 11% for one-year contracts and 3% for two-year contracts.

3.4 Churn by Tenure

Bar chart: Churn is highest among customers with 0-11 months tenure at 48.28% and declines to 6.68% among customers with 60-72 months tenure.

3.5 Churn by Internet Service

Bar chart: Fiber optic customers have 42% churn, compared with 19% for DSL and 7% for customers with no internet service.

3.6 Churn by Payment Method

Bar chart: Electronic-check users have 45% churn, compared with 17% automatic bank transfer, 15% automatic credit card and 19% mailed check.

3.7 Customer Tenure Distribution

Histogram: The distribution shows concentration at shorter and longer tenure ranges, with a notable concentration in the first 12 months and another at 60-72 months.

3.8 Monthly Charges Distribution

Histogram: Monthly charges are spread across a broad range, indicating substantial variation in the amounts paid by customers.

3.9 Monthly Charges by Churn Status

Box plot: Churned customers generally show higher monthly charges than customers who stayed, consistent with the higher average monthly charge.

3.10 Correlation Heatmap

Correlation heatmap: Tenure has the strongest observed relationship with churn at -0.3522. Monthly Charges has a weak positive correlation of 0.1934, while Total Charges has a weak negative correlation of -0.1983.

4. Business Insights

Insight 1 — Contract type

Month-to-month customers represent 55.02% of the customer base and have a 43% churn rate, compared with 11% for one-year contracts and 3% for two-year contracts.

Insight 2 — Tenure

Customers in their first 12 months have a 48.28% churn rate, compared with 6.68% among customers with 60-72 months of tenure. Tenure and churn have a -0.3522 correlation.

Insight 3 — Internet service

Fiber optic customers have the highest observed churn at 42%, compared with 19% for DSL and 7% for customers without internet service.

Insight 4 — Payment method

Electronic-check customers have a 45% churn rate, substantially higher than the other payment methods analysed.

Insight 5 — Monthly charges

Churned customers have an average monthly charge of R76.28 compared with R60.81 for customers who stayed. The correlation between monthly charges and churn is weakly positive at 0.1934.

5. Business Risks

Risk 1 — Customer retention risk

High churn in several important customer segments may reduce recurring revenue and increase the effort and cost required to replace customers.

Risk 2 — Revenue risk

Churned customers have higher average monthly charges than retained customers, creating a potential risk that relatively higher-value recurring revenue is being lost.

Risk 3 — Early-life retention risk

The 48.28% churn rate among customers in their first 12 months indicates a significant risk of losing customers before long-term loyalty is established.

6. Business Opportunities

Opportunity 1 — Early-tenure retention programme

Introduce structured onboarding and proactive engagement during the first 12 months.

Opportunity 2 — Contract conversion

Encourage suitable month-to-month customers to move to longer-term contracts using targeted value propositions and incentives.

Opportunity 3 — Risk-based segmentation

Combine multiple churn-risk indicators to prioritise customers for targeted retention interventions.

7. Business Recommendations

#	Recommendation	Target	Success measure
1	Launch a structured first-year onboarding and retention programme.	0-11 month customers	Reduce first-year churn from the observed 48.28%.
2	Introduce targeted incentives for suitable month-to-month customers to move to longer contracts.	Month-to-month customers	Reduce the observed 43% churn and increase longer-term contract uptake.
3	Investigate the drivers of fiber customer churn, including pricing, service quality, complaints and customer experience.	Fiber optic customers	Reduce the observed 42% churn.
4	Investigate payment friction and encourage suitable electronic-check users to adopt automatic payment methods.	Electronic-check customers	Reduce the observed 45% churn.

5	Develop a monthly churn-risk dashboard to monitor high-risk segments and intervention outcomes.	Management / retention teams	Track churn trends and retention effectiveness monthly.
---	---	------------------------------	---

The recommendations should be evaluated through measurable retention KPIs. The analysis identifies associations rather than causal relationships; interventions should therefore be tested and monitored rather than assuming that any single characteristic directly causes churn.

8. Conclusion

The analysis indicates that customer churn at ABC Communications is concentrated among specific customer segments rather than being evenly distributed across the customer base. Month-to-month contracts, early tenure, fiber optic service and electronic-check payment are associated with substantially higher observed churn. ABC Communications should therefore adopt a targeted retention strategy supported by ongoing data monitoring. Early-tenure engagement, contract conversion, service investigation, payment optimisation and a churn-risk dashboard provide practical starting points for reducing customer losses and protecting recurring revenue.

Appendix A — Key Analytical Metrics

Metric	Result
Overall churn	26.54%
Month-to-month share	55.02%
Month-to-month churn	43%
One-year churn	11%
Two-year churn	3%
0-11 month churn	48.28%
60-72 month churn	6.68%
Fiber optic churn	42%
Electronic-check churn	45%
Average monthly charge — stayed	R60.81
Average monthly charge — churned	R76.28

Tenure ↔ Churn	-0.3522
Monthly Charges ↔ Churn	0.1934
Total Charges ↔ Churn	-0.1983